1. Find the least upper upper bound (if exists) and the greatest lower bound (if it exists).

   (a) $\{x : |x-1| < 2\}$.

   (b) $\{x : \ln x > 0\}$.

   (c) $\{x : \sin x \geq -1\}$.

   (d) $\{-1, -\frac{1}{2}, -\frac{1}{3}, \cdots\}$.

2. Determine the boundedness and monotonicity of the sequence with a_n, as indicated.

   (a) $\frac{n+3}{\ln(n+3)}$.

   (b) $\frac{(-1)^n}{n}$.

   (c) $\frac{3^n}{(n+1)^2}$.

   (d) $\frac{n^2}{\sqrt{n^3+1}}$.

3. Show that, if $0 < c < d$, then the sequence

$$a_n = (c^n + d^n)^{1/n}$$

   is bounded and monotonic.

4. State whether the sequence converges as $n \to \infty$; if it does, find the limit.

   (a) $\left(1 - \frac{1}{n}\right)^n$.

   (b) $\int_0^n e^{-nx} dx$.

   (c) $\left(|t| + \frac{x}{n}\right)^n, t > 0, x > 0$.

   (d) $\frac{\arctan n}{n}$.

5. Show that $\lim_{n\to\infty} [(n+1)^{1/2} - n^{1/2}] = 0$.

6. Show that $\lim_{n\to\infty} [(n^2+n)^{1/2} - n] = \dfrac{1}{2}$.

7. A sequence $a_1, a_2, \cdots$ is called a Cauchy sequence if

   for each $\epsilon > 0$ there exists a index k such that

$$|a_n - a_m| < \epsilon \quad \text{for all } m, n \geq k.$$

   Show that

   every convergent sequence is a Cauchy sequence.

   Note: It is also true that every Cauchy sequence is convergent, but that is more difficult to prove. (You do not necessarily prove it in the homework.)

8. (a) Let $a_1, a_2, \cdots$ be a convergent sequence. Prove that

$$\lim_{n\to\infty} (a_n - a_{n-1}) = 0.$$

   (b) What can you say about the converse? That is, suppose that $a_1, a_2, \cdots$ is a sequence for which

$$\lim_{n\to\infty} (a_n - a_{n-1}) = 0.$$

   Dose $a_1, a_2, \cdots$ necessarily converge? If so, prove it; if not, give a counterexample.

9. Show that if $a_n \to 0$ and the sequence with terms b_n is bounded, then $a_n b_n \to 0$.

10. If $a_n \to L$, show that $|a_n| \to |L|$. If the converse true?

11. Let f be a function continuous everywhere and let r be a real number. Define a sequence as follows:

$$a_r, \quad a_2 = f(r), \quad a_3 = f[f(r)], \quad a_t = f\{f[f(r)]\}, \cdots .$$

Prove that if $a_n \to L$, then L is a fixed point of $f : f(L) = L$.

12. Calculate

(a) $\lim\limits_{x\to\infty} \dfrac{\ln x^k}{x}$.

(b) $\lim\limits_{x\to\infty} \left(1 + \dfrac{a}{x}\right)^{bx}$.

(c) $\lim\limits_{x\to\infty} (x^3+1)^{1/\ln x}$.

(d) $\lim\limits_{x\to 0} (x \ln|\sin x|)$.

(e) $\lim\limits_{x\to\infty} \left(x \sin \dfrac{\pi}{x}\right)$.

(f) $\lim\limits_{x\to 1} \left(\dfrac{1}{\ln x} - \dfrac{x}{x-1}\right)$.

(g) $\lim\limits_{n\to\infty} \dfrac{1}{n} \ln \dfrac{1}{n}$.

(h) $\lim\limits_{n\to\infty} \dfrac{\ln n}{n^p}$, $p > 0$.

(i) $\lim\limits_{x\to 1} \dfrac{\ln x}{1-x}$.

(j) $\lim\limits_{x\to 0} \dfrac{\sqrt{a+x} - \sqrt{a-x}}{x}$.

(k) $\lim\limits_{x\to 0} \dfrac{\arctan x}{\arctan 2x}$.

(l) $\lim\limits_{x\to 0} \dfrac{\arcsin x}{x}$.

(m) $\lim\limits_{x\to 0} \dfrac{a^x - (a+1)^x}{x}$.